

Sai Kumar Bella

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SUMMARY

Founding engineer at Workcuit AI building and operating **StayCuit** — a multi-tenant hospitality SaaS live across 3 production partner properties — end-to-end across native **iOS (Swift / SwiftUI)**, **Angular 17 PWA**, **Node.js / Express backend**, **Firestore data**, and a sole-authored **Chrome Manifest V3 PMS-integration extension**. Built an applied-AI assistant for property owners and managers — multi-agent orchestration with custom **MCP servers**, cost-aware model routing across **Claude Opus 4.7** and self-hosted **NVIDIA NemoClaw / Nemotron**, plus grounding skills for hallucination detection. Prior J2EE / Spring Boot enterprise full-stack at TCS. M.S. Computer Science, with nearly 4 years of hands-on software engineering experience across iOS, web, backend, and applied AI.

EDUCATION

Purdue University Northwest

Master of Science in Computer Science

Hammond, IN

Jan. 2023 – Dec. 2024

Sri Venkateswara Institute of Technology

Bachelor of Technology in Computer Science and Engineering

Andhra Pradesh, India

Aug. 2017 – May 2021

EXPERIENCE

Software Engineer, Founding Team

May 2025 – Present

Workcuit AI, LLC

Illinois (Remote)

- Founding engineer and majority author of the StayCuit monorepo (88 of 119 commits since Jan 2024 across iOS, Android, web, backend, and Chrome extension); reviewed and merged team PRs from 5 other contributors.
- Designed and built the Express.js + Firestore production backend (~30 REST controllers and ~40 services on Node 16+, hosted on Render) including idempotency middleware, App Check integration, structured logging, and rate-limited PMS-credential storage encrypted with **AES-256-GCM**.
- Authored an event-registry orchestration engine that consolidates scattered Firestore writes — condition-based handlers fan out room-status, assignment, and notification side effects from a single dispatch surface.
- Designed the multi-tenant Firestore data model — `‘/properties/{pid}/{rooms,stays,members,devices,events,...}’` with **collectionGroup** queries keyed by property + business date for cheap per-staff fan-out.
- Implemented **custom-claim-based RBAC** across six roles (frontdesk, admin, manager, housekeeping / housekeeping manager, maintenance, owner) enforced both at the API layer and in Firestore security rules.
- Built **PIN-paired custom-token device auth** for shared iPads — server-held HMAC secret, per-device and per-property rate limits in 15-minute windows, and Firebase custom claims scoping every request to (property, role, device).
- Authored the **PMS sync engine** — a three-channel pipeline that reconciles property state from CSV uploads, **Microsoft Graph / Outlook** email parsing, and a real-time Chrome-extension event channel (1500+ LOC, configurable per property).
- Designed a per-property **action policy registry** that resolves PMS diffs against StayCuit state — automatic for safe deltas, human-in-the-loop (`‘ask_fd’`) for ambiguous cases like stayover-vs-new-guest or mistaken checkouts.
- Built the unified scheduling system — shift templates, weekly publish workflow, swap requests, PTO balances with accrual, and **geofenced clock-in/clock-out via haversine-distance** enforcement against property boundaries.
- Implemented a property-aware **night-audit pre-check and day-end rollover** that switches stay defaults to the property’s business date and adjusts derived metrics across stays, shifts, and housekeeping queues.
- Built snapshot-driven **real-time sync between shared iPads and the frontdesk console** — Firestore `‘onSnapshot’` listeners with per-session state machines, eliminating the polling layer the team previously used.
- Designed and built the **Angular 17 + PrimeNG 17 PWA**, role-aware and lazy-loaded by module — frontdesk, housekeeping, manager, maintenance, requests, admin portal, and the guest iPad self-service flow (service-worker offline support, iOS-specific safe-area handling).
- Built the **iPad guest-pad self-service kiosk** — barcode-based ID scanning, returning-guest auto-fill, configurable per-property step flow, and signed reg-card PDFs via `‘signature_pad’` + `‘pdf-lib’`.
- Built the frontdesk single-page check-in flow — stepper-driven UI with a state machine, sticky action bar, optimistic transitions, and a quick ID-scan dialog that combines PDF417 decoding with auto-cropped photo capture in a single transactional save.
- Ported the entire frontdesk + housekeeping + manager PWA to **native iOS in SwiftUI** (~96 files, MVVM with `‘@Published’` / `‘ObservableObject’`, `async/await` `‘APIClient’`, Firestore `‘collectionGroup’` listeners), currently in **TestFlight beta across 3 production partner properties**.
- Engineered **geofenced staff time-tracking on iOS** using **CoreLocation** (CLLocationManager, CLCircularRegion, region monitoring) — restricts clock-in/out to property boundaries, handles iOS permission flows, background monitoring, accuracy-vs-battery tradeoffs, and edge cases like GPS drift and permission revocation.

- Maintained and extended **Objective-C** modules within a mixed Swift/Obj-C codebase — bridging headers, Swift-Obj-C interop, memory management (ARC, retain cycles, weak/strong), and Grand Central Dispatch concurrency.
- **Sole author** of a Chrome **Manifest V3** extension that integrates StayCuit with **HotelKey PMS** — TypeScript + webpack 5, MutationObserver-based SPA DOM tracking, and the MV3 **offscreen-document pattern** to keep Firebase Auth alive across service-worker restarts.
- Mirrors HotelKey PMS events (room selection, confirmation observed, room moves, stay extensions, checkouts) to the StayCuit backend in real time over `/api/properties/:pid/extension-events`, with manual session-binding UI and a field-synonym dictionary for form auto-fill.
- Built the check-in **OCR pipeline** — PDF417 barcode reading via **ZXing** plus **OpenCV.js** document edge detection and auto-crop — and fixed a long-standing Safari PWA memory leak by reusing canvas / reader / decoder instances across frames.
- Designed and shipped an **applied-AI assistant for property owners and managers** — multi-agent system with an orchestration layer that classifies query complexity, routes simple queries to a small model and escalates complex ones to **Claude Opus 4.7**, live across 3 partner properties.
- Built custom **MCP (Model Context Protocol) servers** wrapping StayCuit features so AI agents can fetch property data on demand through scoped, permissioned tool calls — instead of dumping the database into context.
- Deployed **NVIDIA NemoClaw** on dedicated per-property VPS — open-source self-hosted agent stack on Nemotron — so token-heavy report queries are answered from pre-computed reports indexed by metadata descriptions, with policy-based privacy guardrails per tenant.
- Built **grounding / hallucination-detection skills** that verify each extracted datum traces back to the cited source, rejecting agent outputs that can't be attributed to property data.
- Authored a **daily property briefing agent** that summarizes the previous day's shift activity, work logs, and issues for owners and managers — pulls from the scheduling and event-registry data and posts a structured action list each morning.
- Established engineering conventions used by the team — strict conventional-commit messages (`'feat(scope):'`, `'fix(scope):'`, `'perf:'`, `'refactor:'`), a versioning script wired into pre-build/pre-start, CI on **GitHub Actions** (Firebase Hosting) plus **Render** deploys for the API.

Java Full Stack Developer

Aug. 2021 – Oct. 2022

Tata Consultancy Services

Bangalore, India

- Developed enterprise web applications using **Java, J2EE, Spring, and Hibernate ORM**; designed and integrated **REST and SOAP APIs** between internal services and third-party systems on a high-volume production platform.
- Built responsive frontend interfaces with **Angular, JavaScript, HTML, and SCSS**; tuned MySQL through query optimization, indexing, and stored procedures; achieved comprehensive JUnit test coverage in Agile sprints.

Software Engineering Intern

May 2019 – Dec 2020

Sri Venkateswara Institute of Technology

Andhra Pradesh, India

- Developed a role-based campus communication Android app in Java, adopted by 700+ students across 5 departments, with distinct student and faculty flows using Firebase Authentication for signup/login and Firebase Realtime Database for live content sync.
- Built a course-materials module enabling faculty to upload lecture notes and previous question papers as PDFs to Firebase Storage, organized by CSE branch and year, with an in-app PDF picker, upload-progress tracking, and a built-in viewer.
- Implemented campus newsfeed and events features backed by Firebase Realtime Database, rendering live feeds via FirebaseUI FirebaseRecyclerViewAdapter bound to RecyclerViews, with Material Design components (CardView, tabbed fragments).

PROJECTS

StayCuit iOS App | *SwiftUI, Objective-C, Combine, async/await, Firebase*

- Native iOS app for the StayCuit hospitality SaaS — SwiftUI MVVM across ~96 files, Combine, async/await, and Firestore real-time listeners; ported from the Angular PWA and currently in TestFlight beta across 3 partner properties.
- Supports geofenced operations via CoreLocation, shared-device session handoff, multi-role access (frontdesk / housekeeping / manager), and offline tolerance on the property floor.

StayCuit AI CCTV Monitoring (Prototype) | *Node.js, Vision LLMs, OpenCV, Firebase*

- Prototype pipeline for property-wide camera monitoring (60+ cameras per property) using vision LLMs for event detection and natural-language search.
- Active R&D on a cost-effective model strategy given high-volume continuous visual feeds; modular hooks for event-driven Firestore triggers and mobile alerting once a viable model is selected.

College Android App | *Java, Android SDK*

- Android application for college students and professors to share class materials, view on/off-campus notifications, and post to a campus newsfeed.

Result Automation | *Python, Selenium*

- Selenium-based automation that scrapes the university results portal and exports per-student records into an Excel workbook.

Gesture-Controlled Gaming | *Python, OpenCV, NumPy*

- Computer-vision pipeline using OpenCV + NumPy that maps hand gestures and color zones to simulated keyboard input, driving a car-racing game in real time.

Infant Danger Detection | *Python, OpenCV*

- Webcam-based tracking that lets a caregiver mark danger zones in a room and triggers an audible alert when the tracked infant moves near them.

Weekly Management System | *Java*

- Java application for weekly event scheduling with participant invitations and personal calendar organization.

Farm Animal Intrusion Detection | *Python, OpenCV, Selenium*

- OpenCV-based motion detection for farm cameras; on intrusion, captures snapshots and notifies the owner via WhatsApp Web through a Selenium-driven send flow.

Santa Maria Bakery Website | *TypeScript, Web*

- Order-management web app for a local bakery in Hammond, IN, supporting both customer-facing ordering and an owner-side management view.

Social Media OSINT Analysis | *Python, OSINT*

- Open-source intelligence pipeline over public social-media data: threat analysis, user-distribution statistics, and trending-topic surfacing.

Branch Predictor Comparison | *C++*

- C++ simulation comparing Global-Address (GAp) and Per-Address (PAP) branch-prediction strategies; evaluates accuracy and memory cost with a 15-bit Branch History Register and 2-bit Pattern History Table.

HotelKey PMS Chrome Extension | *Manifest V3, TypeScript, webpack 5, Firebase Auth (offscreen)*

- Sole-author Chrome Manifest V3 extension that integrates StayCuit with HotelKey PMS — uses the offscreen-document pattern to keep Firebase Auth alive across service-worker restarts, plus a single master MutationObserver + 1s URL poll to track HotelKey's SPA cheaply.
- Mirrors PMS events (room selection, confirmation, room moves, stay extensions, checkouts) to the StayCuit backend in real time, captures reg-card PDFs from the PMS DOM, and exposes an in-page binding UI plus a field-synonym dictionary for manual form-filling.

StayCuit AI Assistant Platform | *Custom MCPs, Multi-agent orchestration, Claude Opus 4.7, NVIDIA NemoClaw, Nemotron*

- Multi-agent applied-AI system for property owners and managers — orchestration layer classifies query complexity and routes between small cost-effective models and Claude Opus 4.7; live across 3 partner properties.
- Custom MCP (Model Context Protocol) servers wrap StayCuit features so agents fetch property data through scoped, permissioned tool calls instead of context-stuffing the database.
- Deployed NVIDIA NemoClaw on per-property VPS so token-heavy report queries run against pre-computed reports indexed by metadata; policy-based privacy guardrails per tenant via NVIDIA OpenShell.
- Grounding skill verifies each extracted datum traces back to its cited source; a daily property briefing agent summarizes shift activity, work logs, and issues for owners each morning.

TECHNICAL SKILLS

iOS: Swift, SwiftUI, UIKit, Objective-C, Combine, async/await, GCD, CoreLocation, MapKit, URLSession, Core Data, XCTest, Xcode

Languages & Mobile: Swift, Objective-C, Java (Android SDK), JavaScript, TypeScript, Python, SQL, Material Design

Backend & Cloud: Node.js, Spring Boot, J2EE, Hibernate, REST, gRPC, OAuth/JWT, Firebase (Firestore, Auth, Functions, Storage), Google Cloud, CI/CD

AI/ML: LLM Integration, Gemini Vision API, Anthropic Claude, RAG, Prompt Engineering

Architecture & Practices: Multi-tenant SaaS, MVVM, Event-driven, Real-time sync, Location-based services, TDD, Code/Design Review, Accessibility